MATH 110 — Week 2

Homework 2 — Solutions

HW2 P1.

Idea. Choose $w_1 = -v_1$ with $v_1 \neq 0$. Both one-vector lists are independent, but their sum is zero.

1. In $V = \mathbb{R}$, choose the one-element lists $v_1 = 1$ and $w_1 = -1$. Each is independent since a nonzero scalar in a field has an inverse.
2. Their termwise sum is the list (0), which is dependent because $1 \cdot 0 = 0$ has nonzero coefficient. The assertion is false.

HW2 P2(a).

Idea. Each vector-space identity is a field identity at every coordinate.

1. Addition and scalar multiplication produce sequences in F^∞, so the operations are closed.
2. Addition is commutative and associative coordinatewise. The zero sequence is an additive identity and the negative sequence is the additive inverse.
3. For scalars a, b and sequences x, y, each coordinate satisfies $(ab)x_j = a(bx_j)$, $a(x_j + y_j) = ax_j + ay_j$, $(a + b)x_j = ax_j + bx_j$, and $1x_j = x_j$. Thus all vector-space axioms hold.

HW2 P2(b,c).

Idea. Arbitrarily many independent coordinate sequences rule out finite dimension, but finite combinations do not produce every sequence.

1. For each N, the list $e_1, \ldots, e_N$ is independent: its zero combination has jth coordinate equal to its jth coefficient.
2. If a list of length M spanned F^∞, the independent list of length $M + 1$ would contradict Notes Proposition 10. Hence the space is infinite-dimensional.
3. Any finite combination of coordinate sequences is zero past the largest index used. The sequence $(1, 1, \ldots)$ is not, so the set of all e_j does not span.

HW2 P3.

Idea. A basis extracted from vectors in the union remains in that union.

1. By Proposition 14, the subspaces U, W have finite bases, since V is finite-dimensional.
2. Join these bases. The joined list spans $U + W = V$: expand the two summands separately.
3. Proposition 9 deletes vectors to leave a basis. Every retained vector still lies in U or W, proving the assertion.

HW2 P4(a).

Idea. The only dimension strictly between 0 and 2 is 1.

1. The subspace is nonzero, so a nonzero v gives an independent one-element list, and its dimension is at least 1.
2. Its dimension is at most 2 by Proposition 14. Equality would make it all of F^2, contradicting properness; hence its dimension is 1.
3. Any nonzero v is then a basis by Proposition 13, so the subspace is $\operatorname{span}(v)$.

HW2 P4(b).

Idea. A dependence between two vectors can be solved for one in terms of the other.

1. If v, w are a basis and $v = aw$, then $v - aw = 0$ is a nontrivial relation, contradiction. The same applies if $w = av$.
2. Conversely assume neither is a scalar multiple of the other. Neither can be zero, since zero is a scalar multiple of every vector.
3. If $av + bw = 0$ nontrivially, $a \neq 0$ gives $v = -(b/a)w$; if $a = 0$ then $b \neq 0$ gives $w = 0v$. Both contradict the assumption.
4. The list is independent of length $2 = \dim F^2$, so Proposition 13 makes it a basis.

HW2 P5.

Idea. The dimension formula subtracts the intersection dimension. The given dimensions already add to the dimension of the sum, so that subtracted term must be zero.

1. Proposition 16 gives $\dim(U \cap W) = 3 + 5 - 8 = 0$.
2. A zero-dimensional space is $\{0\}$, because its empty basis spans only zero.
3. The intersection is zero and the sum is $\mathbb{R}^8$, so the definition of direct sum gives the conclusion.

HW2 P6(a).

Idea. Evaluation and differentiation preserve the specified zero equalities.

1. The zero polynomial satisfies both conditions.
2. If $p, q \in U$ and $a \in \mathbb{R}$, then $(ap + q)(1) = ap(1) + q(1) = 0$, and $(ap + q)'(1) = ap'(1) + q'(1) = 0$.
3. Thus $ap + q \in U$, proving the subspace property by Week 1's test.

HW2 P6(b).

Idea. Expand in powers of $x - 1$: evaluating the value and derivative at 1 reads off the first two coefficients.

1. Example 18 gives the unique expression $p(x) = c_0 + c_1(x-1) + \cdots + c_4(x-1)^4$.
2. Evaluation at 1 gives $p(1) = c_0$; differentiate and evaluate to get $p'(1) = c_1$.
3. Thus $p \in U$ exactly when $c_0 = c_1 = 0$. The list $(x-1)^2, (x-1)^3, (x-1)^4$ spans U and is independent, as a sublist of that basis.
4. It is a basis with three members, so $\dim U = 3$. The coefficients give the dimension directly.

Exam practice, Question 1(a).

Idea. Show the four generators are independent, then use the dimension-four basis test.

1. A relation $au_1 + bu_2 + cw_1 + dw_2 = 0$ gives $a + d = 0$, $a + b = 0$, $b + c = 0$, $c - d = 0$.
2. Thus $a = -d$, $b = d$, $c = -d$, and $-d = d$. Over $\mathbb{R}$, this implies $d = 0$, hence all coefficients vanish.
3. The four generators are a basis of $\mathbb{R}^4$, proving the sum is all of it. If a vector belongs to $U \cap W$, equating its U and W combinations gives such a relation, so it is zero. The sum is direct.

Exam practice, Question 1(b).

Idea. Solve the same four coordinate equations with the specified target.

1. The equations are $a + d = 2$, $a + b = 1$, $b + c = 3$, $c - d = 0$.
2. The last equation gives $c = d$; the first two give $a = 2 - d$, $b = d - 1$. Thus $2d - 1 = 3$, so $(a, b, c, d) = (0, 1, 2, 2)$.
3. Therefore $u = u_2 = (0, 1, 1, 0)$ and $w = 2w_1 + 2w_2 = (2, 0, 2, 0)$. Their sum is $(2, 1, 3, 0)$, as required.